TOPOLOGY OF A LIMIT OF DECENT ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 1161-1186, following the statement at 1155-1159.

Editorial note. No omissions; the openness of the image W_i is justified by |U_i| -> |X_i| being open.

Allowed standard prerequisites (editorial).

- Criterion for limit topologies. Topology, Lemma topology-lemma-characterize-limit (cross-chapter, cited at ~1162-1163). The three-point criterion for |X| = lim |X_i| as topological spaces.
- Underlying sets of limits of decent spaces. lemma-inverse-limit-sets (same chapter, card L26292, cited at ~1164-1165). |X| = lim |X_i| as sets.
- Directed limits with affine transitions exist. lemma-directed-inverse-system-has-limit (same chapter, card L26291, cited at ~1177-1178). U = lim U_i in the category of schemes.
- Neighbourhoods in limits of schemes. Limits, Lemma limits-lemma-inverse-limit-top (cross-chapter, cited at ~1179-1183). Finds E_i with inverse image inside the preimage of V.

Source reference. Original lemma 07SF: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 878–878. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0CUJ: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1093–1093. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0CUF: focused-seeds/geometry/S9-spaces-limits-topology-limit/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0A2Q: focused-seeds/geometry/S9-spaces-limits-topology-limit/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:spaces-limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *With same notation and assumptions as in Lemma 0CUJ we have* $|X| = \lim_i |X_i|$ *as topological spaces.*

Proof. We will use the criterion of Topology, Lemma 0A2Q. We have seen that $|X| = \lim_i |X_i|$ as sets in Lemma 0CUJ. The maps $f_i : X \to X_i$ are morphisms of algebraic spaces hence determine continuous maps $|X| \to |X_i|$. Thus $f_i^{-1}(U_i)$ is open for each open $U_i \subset |X_i|$. Finally, let $x \in |X|$ and let $x \in V \subset |X|$ be an open neighbourhood. We have to find an i and an open neighbourhood $W_i \subset |X_i|$ of the image x with $f_i^{-1}(W_i) \subset V$. Choose $0 \in I$. Choose a scheme U_0 and a surjective étale morphism $U_0 \to X_0$. Set $U = X \times_{X_0} U_0$ and $U_i = X_i \times_{X_0} U_0$ for $i \geq 0$. Then $U = \lim_{i \geq 0} U_i$ in the category of schemes by Lemma 07SF. Choose $u \in U$ mapping to x. By the result for schemes (Limits, Lemma 0CUF) we can find an $i \geq 0$ and an open neighbourhood $E_i \subset U_i$ of the image of u whose inverse image in U is contained in the inverse image of V in U. Then we can set $W_i \subset |X_i|$ equal to the image of E_i. This works because $|U_i| \to |X_i|$ is open. □